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Digital prediction of ancient ceramic images missing areas based on deep adversarial and reverse diffusion

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Ancient ceramics are subjected to natural disasters, temporal erosion, and other environmental impacts. These historical ceramics exhibit particular vulnerability to glaze wear, disc damage, and fragmentation phenomena. Such degradation frequently results in ceramic images with missing texture, blurred patterns and other issues. To address these challenges, a digital prediction for missing areas in ancient ceramic images through a coupled deep adversarial and reverse diffusion method is proposed. Using deep adversarial patterns prediction and reverse diffusion complementary prediction to construct characteristic equations for patterns and complementary prediction in ancient ceramics. Experimental results demonstrate that model achieves a discriminator loss of 0.620, with MSE and LPIPS metrics reaching 0.0137 and 0.0166, respectively. The proposed method enables effective prediction of missing regions in ancient ceramic images, providing valuable insights into the artifacts' historical context and cultural significance. This advancement holds substantial implications for both the preservation and interpretative transmission of cultural heritage.

Ancient Chinese ceramics represent masterpieces of traditional craftsmanship, encapsulating 5000 years of historical continuity and cultural essence within Chinese civilization¹. The significance of ancient ceramics is profound and multidimensional. This significance lies not only in their artistic value but also extends to multiple aspects. These aspects include history, culture, economy, and science and technology². Ancient ceramics have a decorative role and reflect the social culture, esthetic concepts, and artistic level. They depict myths, historical events, or religious beliefs, demonstrating the ancient people's understanding of faith and life. At the same time, the painting of patterns is a reflection of the potter's skill and reflects the progress of technology^{3,4}. The study on ancient ceramic patterns and types contributes to a more comprehensive understanding and appreciation of the development and ancient societies evolution. However, ceramic after thousands of years of time, environmental, human and other factors, there is inevitably lack, wear, cracking, peeling, fading and other phenomena. With ceramics as China's business card on the world stage, people are increasingly aware of the importance of ceramic heritage protection. Ancient ceramic images are one of the main means of recording multi-dimensional information in ancient ceramic patterns⁵. In recent years, technological archeology has flourished^{6,7}. For example, 3D reconstruction techniques are used to reconstruct ancient monuments and establish the virtual 3D artifact database^{8–10}. Machine vision is used for

heritage image data acquisition and extraction^{11–13}. Radar remote sensing technology is used to collect and locate the position of artifacts^{14–16}. Comparative analysis of ancient ceramics is performed by acoustic characterization of cracks¹⁷. Machine vision technology has a wide range of applications for ancient ceramics, data preservation, and feature extraction. Current image processing techniques based on machine vision predominantly address artifact feature extraction and reconstruction. However, their extension to predicting missing areas in ancient ceramics remains a persistent computational challenge. This paper focuses on the pattern loss and the missing of the plate body in ancient ceramics. Based on deep adversarial and reverse diffusion, the patterns prediction model and complementary prediction model are constructed for ancient ceramic images.

With the continuous development of science and technology, machine vision and deep learning techniques are widely used for image prediction and restoration^{18,19}. Wang et al.²⁰ addressed the problem existing state-of-the-art image restoration methods do not apply to artistic images. A novel style-directed two-branch repair network (SDI-Net) was proposed. The style reconstruction branch (SR) was constructed to provide intermediate supervision for the style repair branch (SI). The restoration results were ultimately obtained by fusing restoration style features with restoration content features. Liao et al.²¹ identified a problem in cross-modal image restoration methods: the lack of relationships between the damaged part of

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the image, the remaining part, and non-visual modalities. To address this, a haptic image restoration scheme was proposed. Feature extraction and cross-attention-based feature indexing were constructed, to find the correspondence between the damaged part of the image and the tactile modality. The damaged portion of the formation is eventually combined with the remaining portion. Ma et al.²² analyzed the poor results of large defect restoration caused by insufficient image guidance. To address this issue, an uncertainty-aware adaptive feedback network (U2AFN) was proposed. An adaptive feedback architecture was developed, progressively fine-tuning repair areas through dynamic integration of multi-scale features. Specifically, hierarchical feedback blocks were designed to propagate high-confidence pixel reconstructions. This mechanism thereby calibrated subsequent optimization iterations through uncertainty-aware guidance. To address the issue of image restoration in the context of noisy images, Wan et al.²³ proposed a nonlocal variational method based on image blocks. The new regularization term was formulated based on the log-likelihood function of the mixed model. The adaptive weight function is built in the desired step, a framework is constructed by it for nonlocal image restoration under noise. Simultaneous image restoration and denoising were finally implemented. To sum up, the above scholars have proposed highly adaptable solutions for image prediction and restoration. These solutions were developed in response to the problems of extensive image corruption and image noise. However, in response to the problem of missing patterns and missing disks in ancient ceramic images with complex patterns. The methods of the above scholars have obvious limitations, and there are still many problems for the restoration of ancient ceramic images.

To achieve pattern prediction and complementary prediction for ancient ceramic images. A digital prediction of missing areas in ancient ceramic images based on deep adversarial and reverse diffusion coupling methods is proposed. The split binocular image acquisition and processing system of ancient ceramics is built and optimized independently. Through the focus adjustment, missing area positioning and other steps to establish the ancient ceramic heritage image dataset. The dataset is used to complete the prediction of missing areas in ancient ceramic images via the pattern prediction model and the complementary prediction model. For the pattern model, adaptive dual-threshold Canny edge detection is utilized to precisely capture image features. Lipschitz continuity constraints are then established to ensure the consistency and smoothness of the predicted areas. Furthermore, spectral normalization and local patch adversarial models are designed. These specifically enhance the model's adaptability to the unique characteristics of ancient ceramic images. In the complementary model, a shift window converter is constructed to facilitate the integration of contextual information. A hierarchical structure, incorporating a shift window mechanism, is developed to capture multi-scale features. Additionally, a Markov chain is employed for reverse diffusion to iteratively refine and generate the predicted images. Prediction of missing areas in ancient ceramic images provides archeologists and historians with more complete information. This helps to understand more accurately the historical background and cultural connotations of the ancient ceramics. Virtual prediction by machine vision reduces labor and material consumption and provides guidance for physical repairs. To a certain extent, it makes up for the losses caused by historical reasons and has far-reaching significance for the protection and transmission of cultural heritage.

Methods

Ancient ceramic image acquisition system

To fully capture features in ancient ceramic images, the split binocular image acquisition and processing system is independently constructed and optimized, and the ancient ceramics database is established. The system consists of three main parts: the image acquisition platform, the process of image acquisition, and the establishment of an ancient ceramic database. The details are shown in Fig. 1.

Figure 1a shows the schematic diagram of the split binocular image acquisition and processing system for ancient ceramics. The lens part of the ancient ceramic images acquisition includes a high-

resolution CMOS sensor, negative, shutter, aperture, and lens. The lens can be moved horizontally, vertically, and rotated radially. Ancient ceramics are fixed to the platform. The platform is motorized and can be moved in any horizontal and vertical direction. Figure 1b shows a blue-and-white porcelain plate with lotus petal and floral patterns. The patterned disk is a round shape, the color of the pattern is indigo blue. The missing portion of the pattern appears white, reflecting the base color of the unglazed texture plate. In addition to the lack of pattern, there are pentagonal and triangular disks missing from the upper left and right corners of the patterned disc, respectively. To record missing information, missing texture masks are made. As shown in Fig. 1c, the missing texture mask image is grayscale, missing patterns, and missing disks are indicated in white, and the undamaged texture and background areas are displayed in black. The process of image acquisition for ancient ceramics is as follows. First of all, the pattern board is fixed on the placement platform. The focus is adjusted by moving the lens and the display platform to form a clear image of the ancient ceramics. Secondly, to locate the missing pattern, locate and identify the missing parts of the pattern and body of the disc. The image data of ancient ceramics is recorded by high-resolution CMOS sensors, resulting in the acquisition of ancient ceramics images. Finally, the measurement and flexibility of binocular lenses are utilized, and the complete topography of ancient ceramics is obtained, to preserve the information of ancient ceramics from all angles. Figure 1d shows the bottom morphology of the blue-and-white porcelain plate with lotus petal and floral patterns. The disc has a certain depth, internal depression, and epitaxial leveling. The exterior of the disc has a small amount of pattern, there are deficiencies. The base of the grain plate is yellow, it has a bulge. Mage's acquisition of ancient ceramics serves as the foundation for establishing its image database. The process of establishing an image database of ancient ceramics is shown in Fig. 1e. The first step is sample selection. There are many kinds of ancient ceramics. In addition to patterned plates, there are also intricately shaped ancient ceramics such as bowls, jars, vases, censers, stands, saucer dishes, incense burners, and goblets with high stems. These ancient ceramics feature significant curvature, making it difficult to fully display their patterns within a single plane, posing significant challenges for the prediction of ancient ceramic images. Therefore, remove the ancient ceramics that bend the bottle, the ancient ceramic patterns are distributed on a plane and are to be entered into the image database of ancient ceramics. Figure 1e₁ depicts a blue-and-white Jade-Hu-Chun vase, the bottle is curved, so do not select it. Figure 1e₂ shows a blue-and-white mandarin duck lotus pond floral pattern plate, with flat patterns, to be selected. The second step is to establish the image database of ancient ceramics. Figure 1e₃ shows some of the images in the database, the types of ancient ceramics in the image are mainly patterned plates. The image channel is RGB, the PPI is 72. The types of patterned plates can be categorized into dragon-cloud pattern plates, blue-and-white lotus petal floral pattern plates, and blue-and-white mandarin duck lotus pond floral pattern plates. In conclusion, a system for the acquisition and processing of split binocular images of ancient ceramics is developed, resulting in the retrieval of ancient ceramic images. Finally, the image database of ancient ceramics is established. Perform data amplification and mask generation. To establish the data foundation for the image prediction of ancient ceramics.

The preprocessing of ancient ceramic image

The ancient ceramic images are preprocessed based on the database. The image preprocessing process of ancient ceramics and the image data analysis are shown in detail in Fig. 2.

Figure 2a is a preprocessing of the image dataset in ancient ceramics. The preprocessing includes size normalization, mask generation, and synthetic data. The resolution of the original image of different ancient ceramics is different. There is plenty of background left around the grain. Therefore, the original image of ancient ceramics is cropped, and the dimensions are

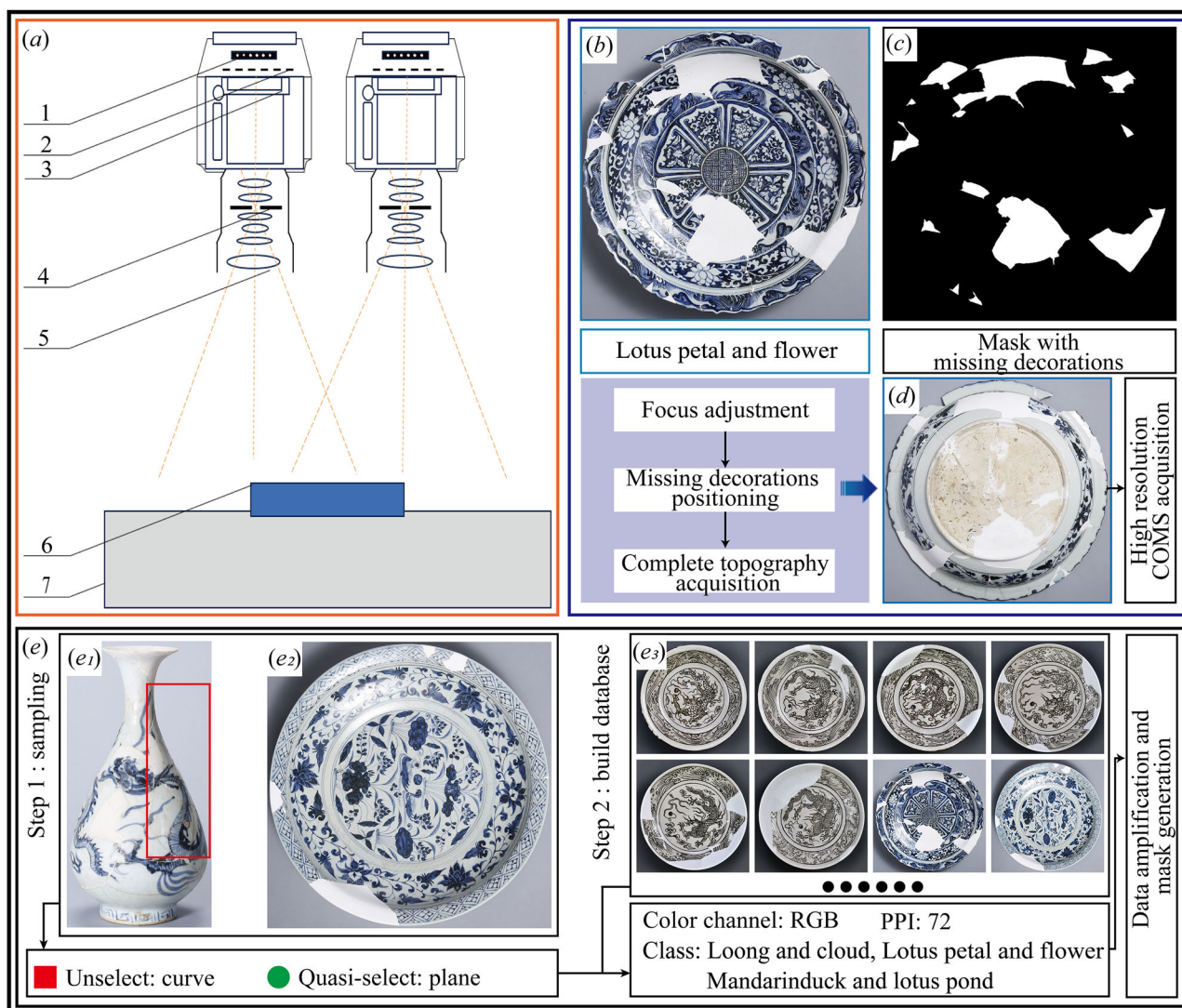


Fig. 1 | Ancient ceramic images acquisition system and database. **a** Image acquisition platform, **b** lotus petal and flower, **c** mask with missing decorations, **d** the bottom morphology of the blue-and-white porcelain plate with lotus petal and floral patterns, **e** the process of establishing an image database of ancient ceramics. 1—High-resolution CMOS sensor, 2—negative plate, 3—shutter, 4—aperture, 5—lens, 6—ancient ceramics, and 7—platform.

normalized. As shown in Fig. 2a₁. The original ancient ceramic image is normalized by size to make the original image of the same size. The edges of the cropped original image are detected and filled, and the mask image is generated by masking the non-missing areas. The mask is a necessary data for the image prediction of ancient ceramics. The mask image records the location of missing patterns and disks of ancient ceramics, scale, and size information, as shown in Fig. 2a₂. Finally, the mask image is combined with the size-normalized ancient ceramic image, and the image dataset of ancient ceramics is generated, as shown in Fig. 2a₃. The composite image retains the size normalization of the pattern in ancient ceramic images without missing parts. The black part of the mask image is removed, the white pattern is missing and the missing parts of the disc are preserved. The absence of images of ancient ceramics is shown in Fig. 2b. The horizontal coordinate represents 9 different images of ancient ceramics, the vertical coordinate represents the proportion of missing areas in 9 different images, and the error bar represents the number of missing areas. The smallest proportion of missing data is observed in Image No. 1, representing a mere 0.12%. The largest proportion of missing areas is observed in Image 6, which accounts for 16.39% of the total area. Conversely, the image contains the fewest number of missing areas, with only 2. Image 8 also exhibits a considerable proportion of missing areas, representing 12.03% of the total area. However, it contains the largest number of missing areas, with 13. In summary, the

image dataset of ancient ceramics is preprocessed, and the missing situation is analyzed. It provides prior data for the expansion of the image dataset of ancient ceramics.

Data augmentation and data distribution in ancient ceramic images

Due to the limited number of ancient ceramic images, the ancient ceramic image database after preprocessed is augmented to form the ancient ceramic image dataset. Among them, amplification mainly relies on rotational means and the introduction of virtual damage. The detailed data augmentation and data distribution are shown in Fig. 3.

Figure 3a₁ shows data augmentation. With the center pixel of the image as the rotation center, the image is rotated. The four corners of the image are cropped to keep the size constant. Meanwhile, the rotated top left, top right, bottom left, and bottom right parts of the image are filled with similar background colors, and the background consistency of the image is maintained. The rotation Angle of the sample image is 141.20°, fill colors are [98, 100, 112]. Figure 3a₂ shows the division of the dataset using MATLAB. The specific measure is to randomly rotate and divide by the 'central rotation' operation described above, with the angle and number of rotations randomized each time. Angle and number of rotations are generated by a 'for loop' combined with a random number function. This is calculated as (total

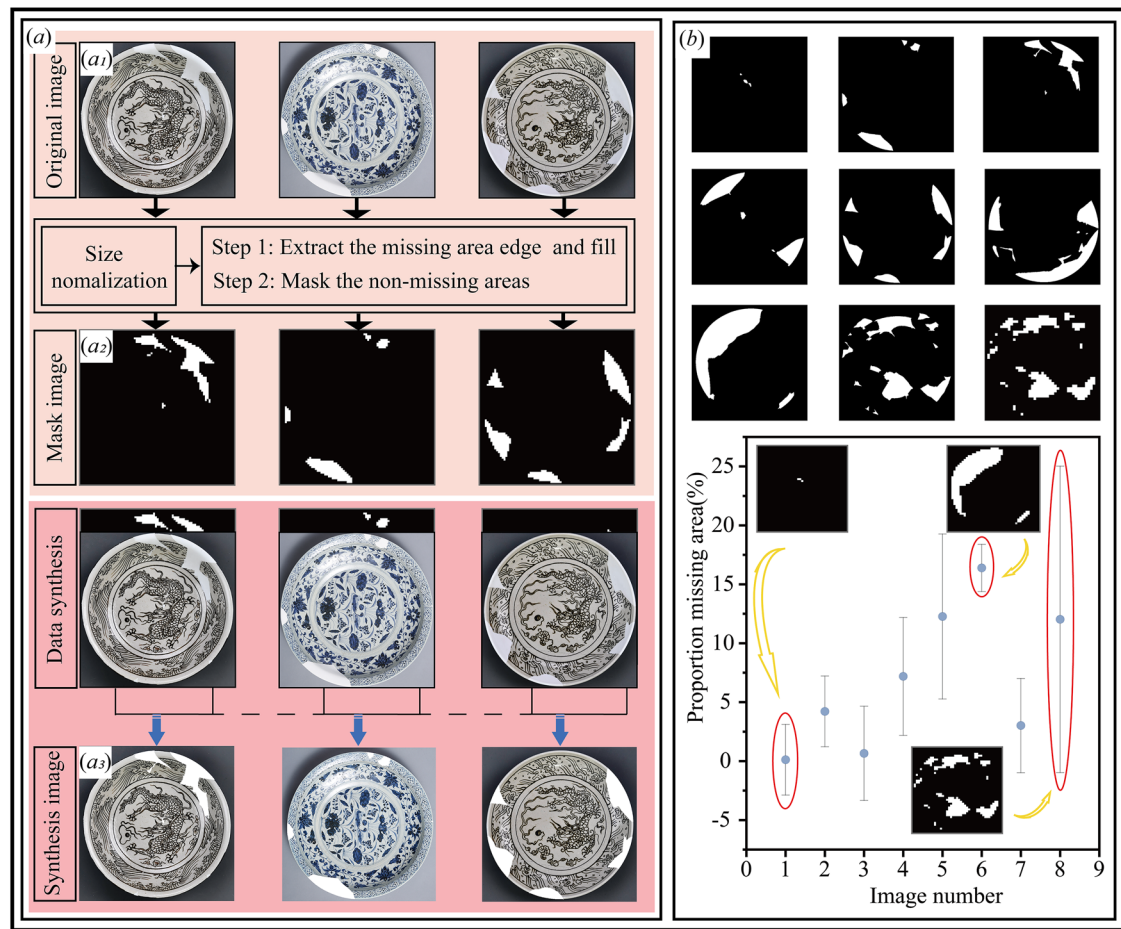


Fig. 2 | The preprocessing process of ancient ceramic images. a The image preprocessing process of ancient ceramics, **b** the image data analysis.

number of images/original number of images)—fixed number + random number. The system parameters are set as follows: total number of real damage pictures 800, virtual damage 200; number of original pictures 8; fixed number 100; random number range [80,120]; rotation angle range [0, 2π]. The actual record shows that in the real damage category, the pictures are rotated 112, 82, 108, 104, 107, 92, 98, and 97 times, respectively. In the virtual damage category, the pictures are rotated 41, 6, 41, 28, 4, 12, 25, and 43 times, respectively. Finally, a certain percentage will be set, and the datasets will be randomly divided into a training set and a testing set. The ratio of the testing set to the training set is approximately 1:7, with a specific number of 128,872. Both the testing set and the training set include actual and virtual damage. The number of actual and virtual damage is 10,226 in the test set and 698,174 in the training set, respectively. The distribution of the expanded image dataset of ancient ceramics is shown in Figs. 2 and 3. The horizontal coordinate of the scatter plot indicates the rotation angle, the range is [0, 360]. The ordinate is the proportion of the missing area, and the range is [0, 0.5]. The color of the scatter indicates the density of the distribution. A red color indicates a high density, whereas a blue color indicates a low density. The true damage dataset is defined as the true pattern missing or disc missing on ancient ceramics. The virtual damage dataset is defined as a virtual irregular graph. In the scatter plot, there are seven green lines, the missing areas accounted for 0.12%, 4.22%, 6.59%, 7.19%, 12.27%, 16.39%, 3.02%, and 12.03%. The real damage dataset is distributed in it, it is a point of rotation with an Angle of zero. The true damage distribution density of rotation is in the middle, it is displayed in green. There are also parts of it that are more densely distributed. For example, in the image of ancient ceramics with a missing area of 16.39%, [100, 120] and [240, 250] have a high density of rotation angles. The distribution range of the missing area ratio of the virtual damage dataset is about [0.0065, 0.4745]. Among them, the

proportion of missing areas is >0.3 . The distribution is mainly concentrated in the range [0.05, 0.25]. The distribution density of the virtual damage dataset is uniform, the scatter color is purple, representing a rotation Angle corresponding to a missing area proportion. Figure 3b is the image of ancient ceramics synthesized by mask. They are dragon-cloud pattern plates, blue-and-white lotus petal and floral pattern plates, and blue-and-white mandarin duck lotus pond floral pattern plates. The rotation Angle is 0, missing areas vary in size, shape, and location. Figure 3c shows a mask image of partial virtual damage. The four types of virtual damage were vertical line, oblique line, arc line, and irregular smear. The missing area accounted for 1.09%, 5.36%, 13.66%, and 25.35%, respectively. The corresponding rotation angles are 120.3°, 276.06°, 239.76°, 277.05°. To sum up, virtual damage and rotational real damage increase the robustness of the data. The ability to predict various sizes, locations, and scales of damage when simulating other forms of deterioration in ancient ceramics.

Digital prediction of ancient ceramic images missing areas by coupling deep adversarial and reverse diffusion methods

For digital prediction of missing areas in ancient ceramic images. The pattern prediction model and complementary prediction model for ancient ceramic images are constructed. The details are shown in Fig. 4.

Figure 4a₁ is the input image of the pattern prediction model for ancient ceramic images. The input image includes the original image of ancient ceramics, the corresponding masked image, and the masked synthesized image. The pattern prediction model is based on generative adversarial networks. The generator network of the pattern prediction model for ancient ceramics is shown in Fig. 4a₂. The generator network consists of encoders, decoders, and residual blocks. The encoder down-samples the input images of ancient ceramics as well as the masked synthetic

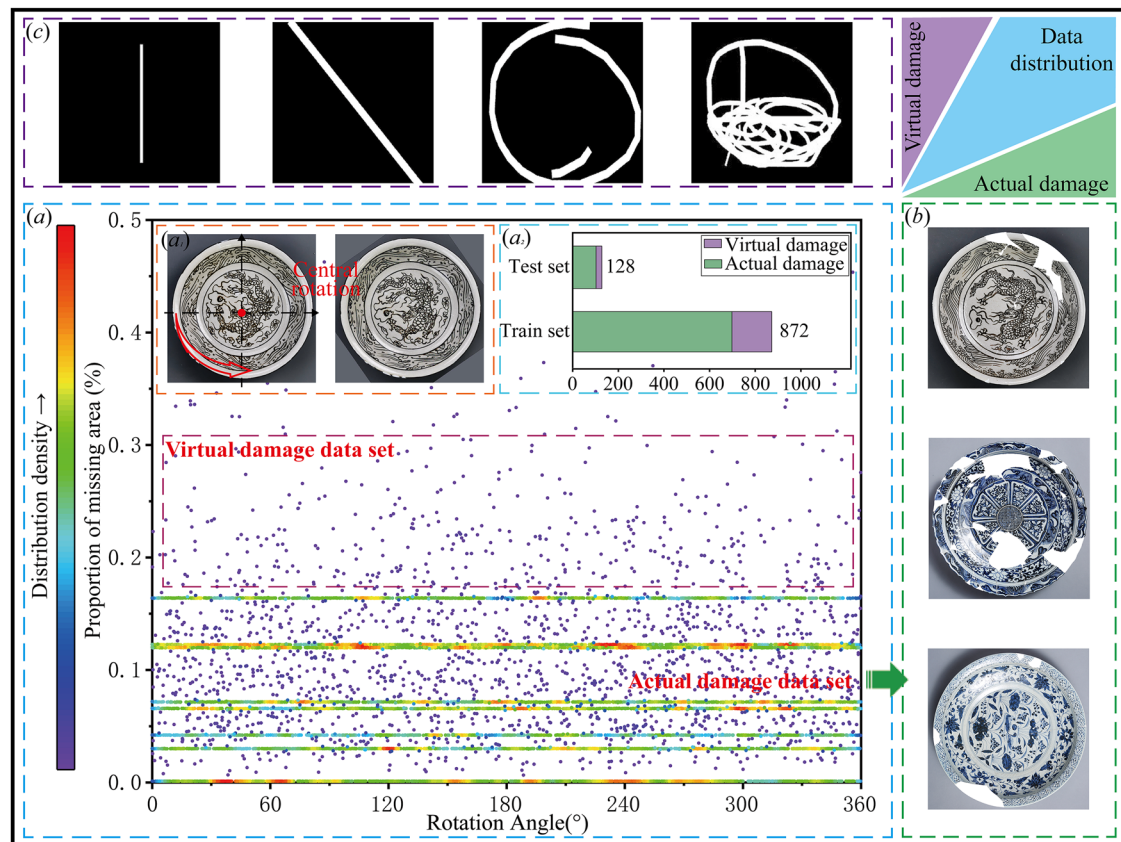


Fig. 3 | Schematic diagram of the data augmentation and data distribution in ancient ceramic images. a Data distribution, **b** actual damage, **c** virtual damage.

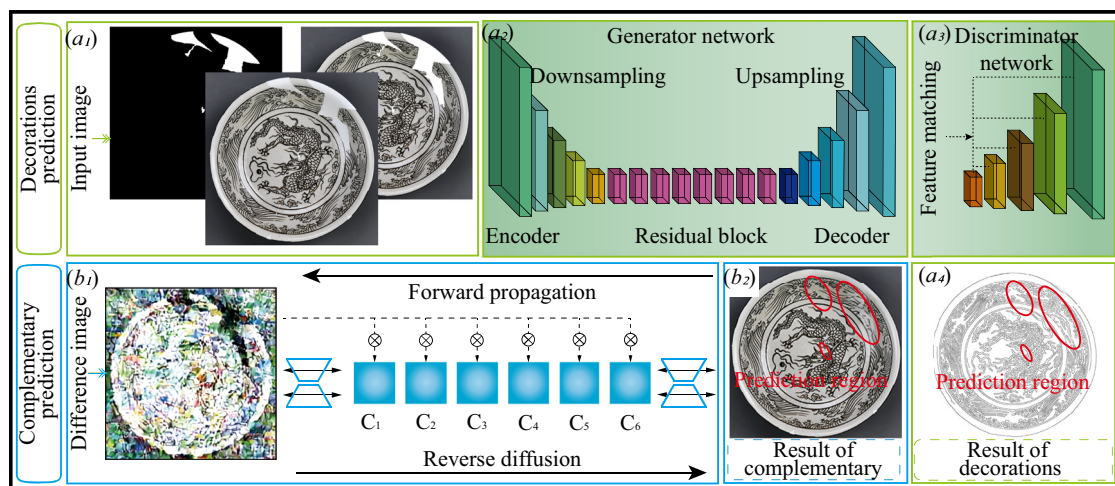


Fig. 4 | Schematic diagram of the digital prediction model for missing areas in ancient ceramic images.

images, generate a feature map. The down-sampling process is implemented using a maximum pooling layer. The amount of data at each level is reduced, thereby reducing computational requirements and helping the model learn more abstract features. The process of down-sampling entails a progressive reduction in the spatial dimension of the image, accompanied by an increase in the number of feature mappings. The residual block is responsible for addressing the issues of gradient vanishing and gradient explosion during the training process of the pattern prediction model. The input to the residual block is denoted as, the output of the main path is denoted as $F(r)$. The output of the residual block is the sum of these two, it is $r + F(r)$. The main path learns how to transform the input r into the residual of the output

$F(r)$. In the decoder, the up-sampling process is implemented using an inverse convolution, increasing the amount of data in each tier. At the same time, up-sampling gradually increases the spatial dimension of the image, and reduces the number of feature mappings. In a network of discriminators, feature matching changes the target of the generator. The output of the discriminator's intermediate layer for the generated ancient ceramic patterns approximates the features of real samples. The complementary prediction model for ancient ceramics is based on diffusion, as shown in Fig. 4b1. The different image is utilized to guide the training of the complementary prediction model for ancient ceramics. The original image to the predicted image is realized by forward propagation. The forward

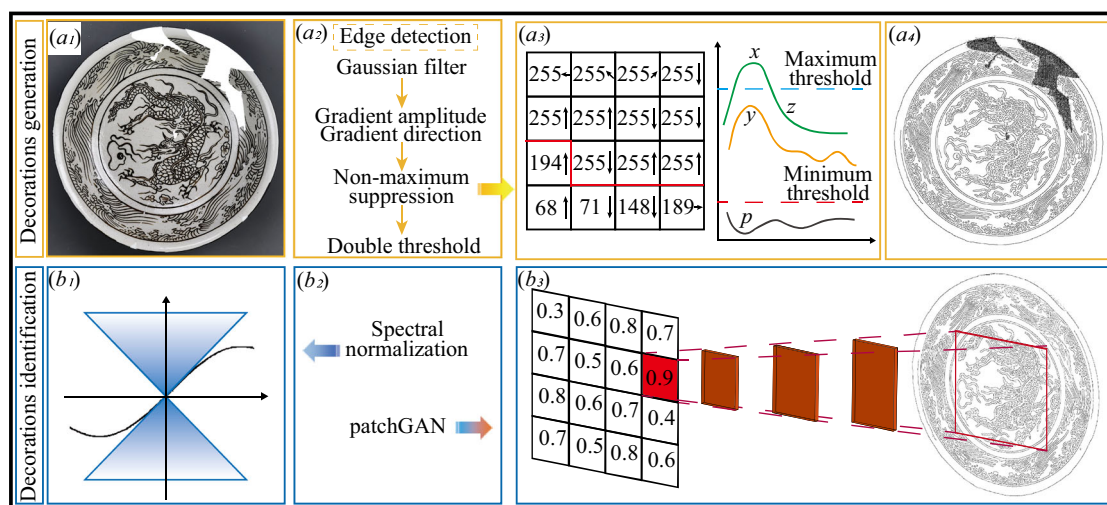


Fig. 5 | Schematic diagram of the pattern prediction model based on deep adversarial.

propagation processes are linked by Markov chains. Defining the original image as R , the predicted image is P , the difference image is D . The difference image D is shifted along the Markov chain, Make the original image migrate R to the predicted image P . $\{C_l\} = \{C_1, C_2, C_3, C_4, C_5, C_6\}$ is an array of shifts, as the step size L (Markov chain length) increases. The pattern prediction model and complementary prediction results of ancient ceramic images are shown in Fig. 4a₄ and b₂, respectively. Of the predicted results of the pattern, the textures in the projected area are connected by black lines. Complementary prediction model with missing areas of pattern filled in. In summary, Combining the decorative and color characteristics of ancient ceramic images. The pattern prediction model and complementary prediction model for ancient ceramics are constructed. The digital prediction for ancient ceramic images is achieved.

The pattern prediction model based on deep adversarial

To achieve the pattern prediction of missing areas in ancient ceramic images, the pattern prediction model is constructed based on deep adversarial. Deep adversarial mainly consists of spectral normalization and local patch adversarial. The model is divided into two modules: pattern generation and pattern identification. The details are shown in Fig. 5.

Figure 5a₁ presents a masked and composited image of the dragon-cloud pattern plate. Figure 5a₂ represents the process of extracting the edge information for ancient ceramics. Firstly, a Gaussian filter is used to smooth the image and suppress noise. The Sobel operator calculates the gradient magnitude and direction of the pixel points. Then, non-extremely large value suppression is performed to refine the edges by setting the maximum threshold and minimum threshold, as shown in Fig. 5a₃. Next, the determination of decorative edges by adaptive dual thresholding, preserving valid edges and suppressing independent weak edges. Finally, the generation of the to-be-predicted patterns is shown in Fig. 5a₄, where the patterns are represented as black lines and the missing parts are shown as black grids. The pattern identification module contains spectral normalization and local patch confrontation. To solve the problem of gradient vanishing caused by the overlap between the distribution of generated data and real data, Lipschitz continuity constraint is introduced for spectral normalization, which makes the neural network training more stable-and-easy to converge. Figure 5b₁ shows the Lipschitz continuity schematic. The curve in the coordinate system is $f(x) = \sin x$ and the maximum slope of the function is 1, called 1-Lipschitz. the blue area is its limiting area. To solve the pattern-blurring problem, the local patch adversarial is designed to replace the discriminator with a fully convolutional network. The input is mapped to a matrix reflecting the probability that a local patch is a true sample, as shown in Fig. 5b₃. In conclusion, it is shown as a pattern prediction model for ancient

ceramic images. Among them, the pattern identification module generates patterns by suppressing weak edges through adaptive dual-threshold Canny edge detection. And the problem of gradient vanishing and pattern blurring is solved by spectral normalization and local patch confrontation.

The complementary prediction model based on reverse diffusion

To realize the complementary prediction of missing areas in ancient ceramic images, a U-shaped network structure is built, shift window converter is embedded. Constructing Markov chains for reverse diffusion, and the prediction of the missing area in ancient ceramic images is realized. The details are shown in Fig. 6.

Figure 6a shows the U-shaped network structure. The network has four layers and contains four up-samples and down-samples. The yellow, purple, orange, green, and blue arrows indicate the ReLU activation function output feature channels, maximum pooling, inverse convolution, convolution, and copy cropping operations, respectively. Input images of ancient ceramics, after which the images are convolved with ReLU activation, and feature maps are generated. Features are down-sampled layer by layer (size halved step by step) by maximum pooling and features are extracted by convolution twice per layer. The result of the VALID convolution is reduced due to the inverse convolution not being able to restore the image at the time of up-sampling. Thus, the down-sampled corresponding layer feature maps are cropped and spliced with the inverse convolution results to compensate for the size loss of the valid convolution. For large resolution requirements, a shift window converter is used instead of a conventional vision converter. That reduces the computational effort while preserving global feature interactions to improve the efficiency and accuracy of color prediction. Figure 6b shows a schematic diagram of the shift window converter. Constructing hierarchical structures to extract multiscale features. A local sliding window mechanism (in-window self-attention computation combined with global sliding overlay) is introduced to reduce computational complexity while preserving global information. Further design of shift window mechanisms across hierarchical levels: alternating standard and shift windows to enhance feature interaction and dependency capturing between different levels of windows. Figure 6c is the shift window converter block structure. A dual module of window polytope attention and shift window polytope attention is designed to replace the traditional standard polytope self-attention. Each module consists of a stack of 'LN layer-attention module-LN layer-MLP', with two modules corresponding to two attention mechanisms. Figure 6d shows the Markov chain schematic. Q_1, Q_2, Q_3, Q_4 denote different random

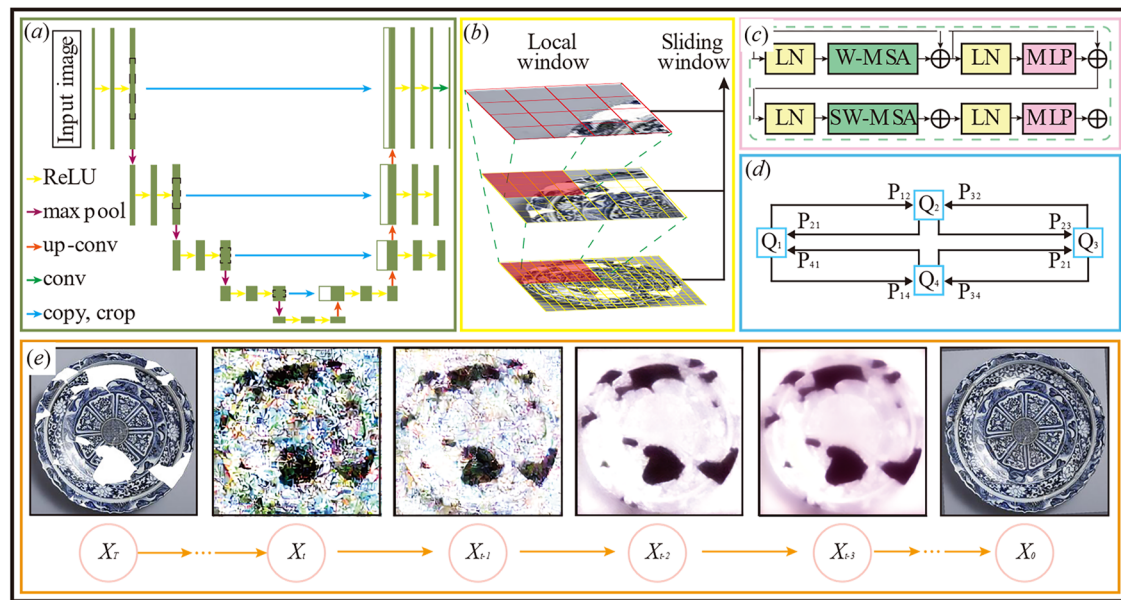


Fig. 6 | Schematic diagram of the complementary prediction model for missing areas in ancient ceramic images based on the diffusion model. a The U-shaped network structure, **b** schematic diagram of the shift window converter, **c** the shift window converter block structure, **d** the Markov chain schematic, **e** schematic diagram of the reverse diffusion process of the complementary prediction model for ancient ceramic images.

events and P_{ij} is the probability distribution of random events Q_i, Q_j . P_{ij} is transformed by a finite number of sequences, eventually converging to a steady state distribution independent of the initial distribution. Figure 6e shows a schematic diagram of the reverse diffusion process of the complementary prediction model for ancient ceramic images. The model is mapped to the latent space via a Markov chain that gradually adds noise to the data X_i at each time step t to obtain the posterior probability $P(X_i; T|X_0)$. The diffusion process is a stochastic process of transitions from one state to another in the state space. The forward diffusion process is as follows, starting from the cyanotype lotus petal floral pattern disc color prediction image X_0 . Gradually adding noise, and finally evolving into the cyanotype lotus petal floral pattern disc masked synthetic image. The reverse diffusion process is as follows, traversing reverses along the Markov chain and training the probability distribution P to generate a new X_0 . In summary, based on a U-structured network, embedding shift window converter and constructing Markov chain for forward propagation. The prediction of the missing pattern and plate parts in ancient ceramic images is achieved.

A mathematical model of pattern prediction for ancient ceramic images based on deep adversarial

The canny edge detection operator returns the gradient amplitude and gradient direction, it is used for non-maximum suppression and adaptive double threshold determination edge. Details are shown in the following formula:

$$\begin{cases} G = \sqrt{\frac{I(x+1,y)-I(x-1,y)}{2}}^2 + \frac{I(x,y+1)-I(x,y-1)}{2}}^2 \\ \theta = \arctan\left(\frac{I(x,y+1)-I(x,y-1)}{I(x+1,y)-I(x-1,y)}\right) \end{cases} \quad (1)$$

In the formula, G is the gradient amplitude, $I(x, y)$ represents the gray value of the pixel point (x, y) , $I(x+1, y)$ and $I(x-1, y)$ represent the gray values of pixels $(x+1, y)$ and $(x-1, y)$, respectively. Among them, horizontal gradient amplitude $G_x = [I(x+1, y) - I(x-1, y)]/2$, the vertical gradient amplitude $G_y = [I(x, y+1) - I(x, y-1)]/2$. The gradient amplitude and gradient direction are the basic conditions for non-maximum suppression and adaptive double threshold to determine the edge. Non-

maximum suppression by going through the pixels one by one, to determine whether the current pixel is the maximum value of the surrounding pixel with the same gradient direction. According to the judgment result, decide whether to suppress this point. Adaptive double threshold According to the gradient amplitude and maximum threshold of the current edge pixel, the relationship between minimum thresholds determines whether the edge is valid, an invalid edge, or a weak edge.

The distribution of the generated data overlaps with the distribution of the real data, so the Jensen–Shannon Divergence is a constant. This eventually causes the gradient of the generator to go to 0, The gradient disappears. Therefore, KKT conditions are used to determine the optimality of the optimal Lipschitz continuous function, a nonlinear programming problem. The KKT conditions are as follows:

$$\begin{cases} \lambda_i \geq 0, \forall i \\ 0 \leq \lambda_i(x - x^*) \leq g_i(x), \forall i \\ \frac{\partial f}{\partial x_j}(x^*) = \lambda_j, \forall j \\ g_i(x^*) \geq 0, \forall i \\ \frac{\partial g_i}{\partial x_j}(x^*) \cdot (x_j - x_j^*) \leq 0, \forall i, j \end{cases} \quad (2)$$

In the formula, λ_i is the weak multiplier associated with the i th constraint, $g_i(x)$ is the i th constraint function, $f(x)$ is the objective function, x^* is the optimal solution, x^{i*} is the equality constraint value of the i th constraint. KKT conditions are optimality conditions for convex optimization problems. They ensure that while all constraints are met, the objective function is locally optimal. In the Lipschitz continuity, For the discriminator $D: I \rightarrow \mathbb{R}$ (I represents the image space). The discriminator is K Lipschitz continuous under the following conditions:

$$\sqrt{\sum_{i=1}^n |D(x_i)|^2} - \sqrt{\sum_{i=1}^n |D(y_i)|^2} \leq C \left[\sqrt{\sum_{i=1}^n |x_i|^2} - \sqrt{\sum_{i=1}^n |y_i|^2} \right] \quad (3)$$

Among them, $\|D(x)\| = \sqrt{\sum_{i=1}^n |D(x_i)|^2}$ is the L^2 norm in the discriminator x direction, $\|D(y)\| = \sqrt{\sum_{i=1}^n |D(y_i)|^2}$ is the L^2 norm in the y

direction of the discriminator, and C is the Lipschitz constant. When C takes the minimum value, The discriminator $D()$ reaches Lipschitz continuity. For images of ancient ceramics, matrix operations have an even greater advantage, the Lipschitz continuity in matrix form is represented as follows:

$$\begin{cases} \langle (M^T M - C^2)x, x \rangle \\ = (M^T M - C^2) \sum_{i=1}^n x_i \cdot \eta_i, \sum_{j=1}^n x_j \cdot \eta_j \\ = \sum_{i=1}^n \sum_{j=1}^n x_i x_j (M^T M - C^2) \eta_i, \eta_j \\ = \sum_{i=1}^n (\lambda_i - C^2) x_i^2 \leq 0 \end{cases} \quad (4)$$

$$\sum_{i=1}^n (C^2 - \lambda_i) x_i^2 \geq 0 \quad (5)$$

In the formula, M is the matrix form of the discriminator D , $M: \mathbb{R}^n \rightarrow \mathbb{R}^m$. The $\langle \rangle$ operator is the inner product operator. The matrix $M^T M$ is a semi-definite matrix, all eigenvalues are non-negative. $\{\eta_1, \eta_2, \eta_3, \dots, \eta_n\}$ is the basis of the matrix $M^T M$, $\{\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n\}$ is the eigenvalue. The mapping $M: \mathbb{R}^n \rightarrow \mathbb{R}^m$ satisfies the Lipschitz continuity condition that the minimum value of C is $\sqrt{\lambda_1}$.

The discriminator contains multiple layers of neural networks, the spectral norm of the diagonal matrix corresponding to ReLU s at most 1. For $f(x)$ to satisfy the Lipschitz constraint, normalization is performed.

$$\begin{aligned} \|\nabla_x(f(x))\|_2 &= \|D_N \frac{W_N}{\max_{\|h\|=1} \|W_N\|_2} \cdots \\ \cdots D_1 \frac{W_1}{\max_{\|h\|=1} \|W_1\|_2}\|_2 &\leq \prod_{i=1}^N \frac{\max_{\|h\|=1} \|W_i\|_2}{\max_{\|h\|=1} \|W_i\|_2} = 1 \end{aligned} \quad (6)$$

In the formula, D_N is a diagonal matrix, used to represent ReLU, when the corresponding input is negative, the diagonal element is 0; When the corresponding input is positive, the diagonal element is 1. W_N is the network parameter matrix. The Lipschitz constraint in multi-layer neural networks is satisfied by dividing the network parameters of each layer by the spectral norm of its parameter matrix, known as spectral normalization.

To solve the problem L1-loss and L2-loss cannot contribute to high-frequency information, resulting in fuzzy patterns. Design local patch countermeasures. Suppose the image is divided into patches of size $N \times N$, the discriminator outputs an $N \times N$ matrix, H_{ij} represents the probability of the patch in row i and column j . Local patch countermeasures consist of the following two steps:

$$F_l = \left(\frac{\text{outputsize}_l - 1}{\text{stride}_l} + 1 \right) \times \left(\frac{\text{outputsize}_l - 1}{\text{stride}_l} + 1 \right) \quad (7)$$

Among them, outputsize_l and stride_l are the output sizes and steps of layer l . The sensory domain F_l represents the area of the image that layer l can cover. The above steps are used to calculate the sensitivity domain of the convolution layer, the next step is to calculate the average of the probability matrix. For the whole image, local patch antagonism calculates the average of the probability matrix for all patches, as the probability of the image being true.

$$P(r) = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N w_{ij} X_{ij} \quad (8)$$

In the formula, $P(r)$ is the probability that the entire image given by the discriminator is the true image. N Indicates the number of patches. w_{ij} is the weight of the patch in row i and column j . Determined by the significance of the patch and the size of the gradient, if a patch contains a distinct edge in the image, then the weight of this patch increases. X_{ij} indicates the probability of

the patch in row i and column j . Precision and Recall indexes were used to measure the model prediction. Details are shown in the following formula:

$$\begin{cases} \text{Precision} = \frac{TP}{TP+FP} \\ \text{Recall} = \frac{TP}{TP+FN} \end{cases} \quad (9)$$

In the formula, Precision and Recall indicate accuracy and recall, respectively. TP means predicting the correct number of pixels, and FP and FN are the number of pixels predicted incorrectly and missed, respectively. The loss function is as follows.

$$\begin{cases} \min_G \max_D L_G = \min_G (\varphi \max_D (M + N)) \\ M = \mathbb{E}_{(V_r, V_p)} [\log D(V_r, V_p)] \\ N = \mathbb{E}_{V_p} \log [1 - D(G(\tilde{V}_p, \tilde{V}_r, M), V_p)] \end{cases} \quad (10)$$

In the formula, G and D stand for generator and discriminator respectively. L_G is against loss, φ is a regularization parameter to prevent overfitting of the model. The symbol $\mathbb{E}$ represents the expected value of the log-likelihood loss function. V_r is the original decorative edge image of ancient ceramics, and V_p is a gray image of ancient ceramic. M is the mask image. $\tilde{V}_p = V_p \odot (1 - M)$ represents the composite grayscale image of the mask of ancient ceramic. $\tilde{V}_r = V_r \odot (1 - M)$ represents the composite image of the original decorative edge mask of ancient ceramic.

A mathematical model of complementary prediction for ancient ceramic images based on reverse diffusion

Shift window converter embedding ensures the high efficiency of prediction, and the performance and applicability of the model are improved.

$$\begin{cases} \Lambda(\text{MSA}) = 4mnD^2 + 2(mn)^2D \\ \Lambda(W - \text{MSA}) = 4mnD^2 + 2B^2mnD \end{cases} \quad (11)$$

In the formula, $\Lambda(\text{MSA})$ and $\Lambda(W - \text{MSA})$ represent the computational complexity of the multi-head attention mechanism and the windowed multi-head attention mechanism, respectively. m and n indicate the length and width of the patch respectively. B is the number of patches, that is, there are $B \times B$ patches in the picture. D is the matrix dimension. The computational complexity difference between the multi-head attention mechanism and the windowing multi-head attention mechanism lies in the second item. Since the number of patches inside the window is smaller than the number of patches in the picture, the number of Windows stays the same. The computational complexity of W-MSA is linearly related to the image size, reducing the computational complexity of the model. The window multi-head attention mechanism reduces computational complexity, but the lack of information exchange between non-coincidence windows. Loss of the ability of the converter to use self-attention mechanisms to construct relationships globally. Therefore, a shift window multi-head attention mechanism is constructed to realize cross-window information exchange. The Markov chain is constructed as follows.

$$P(x_{t+1} | \cdots, x_{t-2}, x_{t-1}, x_t) = P(x_{t+1} | x_t) \quad (12)$$

In the formula, P is a probability, x is a random event. In the Markov chain, the probability of a state transition at a given time depends only on its previous state, the state represented in the formula as x_{t+1} is related only to x_t .

The prediction of ancient ceramic images based on diffusion model is divided into two processes, forward propagation and reverse diffusion, corresponding to denoising and denoising respectively. The forward process cannot directly predict missing images, but this is the basic condition for building a training sample. State x_t is computed at each time step sequentially from the initial state x_0 until the final time T . The state at each time step

depends on the state x_{t-1} at the previous moment, generating the state at each moment and providing the probability distribution of the entire state sequence to assess the quality of the generation of the entire sequence for inference. The probability distribution of forward propagation is shown in Eq. (13).

$$\begin{cases} P(x_t|x_{t-1}) = \rho(x_t; \sqrt{1-\nu_t}x_{t-1}, \nu_t\mathbf{I}) \\ P(x_{1:T}|x_0) = \prod_{t=1}^T P(x_t|x_{t-1}) \end{cases} \quad (13)$$

Among them, x_0 is the real sample, T is the number of times the noise is added, $\{x_1, x_2, x_3, \dots, x_T\}$ is an image containing noise, $\{\nu_t \in (0, 1)\} Tt = 1$ is a hyperparameter of the variance of the Gaussian distribution, used to control step size. Since the forward propagation process occurs along the Markov chain, t is only related to $t-1$. The larger t , x_t the more noise in, $T \rightarrow \infty$, x_T is pure Gaussian noise. The probability distribution of the reverse diffusion process is shown in Eq. (14).

$$\begin{cases} P(x_{t-1}|x_t, x_0) = \exp\left(-\frac{1}{2}(A - B + C(x_t, x_0))\right) \\ A = \left(\frac{\mu_t}{\nu_t} + \frac{1}{1-\mu_{t-1}}\right)x_{t-1}^2 \\ B = \left(\frac{2\sqrt{\mu_t}}{\nu_t}x_t + \frac{2\sqrt{\mu_{t-1}}}{1-\mu_{t-1}}x_0\right)x_{t-1} \\ \tilde{\nu}_t = 1/\left(\frac{\mu_t}{\nu_t} + \frac{1}{1-\mu_{t-1}}\right) = \frac{1-\mu_{t-1}}{1-\mu_t} \cdot \nu_t \\ \tilde{\mu}_t = \frac{1}{\sqrt{\mu_t}}\left(x_t - \frac{1-\mu_t}{\sqrt{1-\mu_t}}\kappa_t\right) \end{cases} \quad (14)$$

In the formula, the three exponents on the right of the first equation are the variances of x_{t-1} . The meaning of x_{t-1} the terms that have nothing to do with x_{t-1} . $\tilde{\nu}_t$ is the variance of x_{t-1} , $\tilde{\mu}_t$ is the meaning of x_{t-1} . κ_t is a hyperparameter. Under the condition given x_0 . The meaning of the posterior Gaussian distribution depends only on x_t and κ_t . The variance is only related to the hyperparameter ν_t .

Experimental environment configuration

The digital prediction of missing areas in ancient ceramic images, achieved through deep adversarial and reverse diffusion, is facilitated, deep learning framework and experimental environment are built. Detailed configurations are shown in Table 1.

Because some of the library files used in the prediction of missing areas in ancient ceramic images through deep adversarial and reverse diffusion are

Table 1 | Experimental environment configuration

Item	Argument
Operating system	ubuntu18.04
CPU	Intel(R) Xeon(R) Platinum 8255C
GPU	V100-SXM2
Deep learning framework	PyTorch1.1.0
Python	3.7
Cuda	10.0
numpy	1.19
scipy	1.0.1
future	0.16.0
matplotlib	2.2.2
pillow	6.2.0
opencv-python	4.6.0
scikit-image	0.14.0
Learning rate	0.0001
Batch size	8

based on Ubuntu. The hardware configuration of the experimental environment is as follows. The operating system is Ubuntu, the version is 18.04. The GPU is a V100-SXM2, it is based on the Volta architecture, providing higher Tensor core performance, and favorable hardware acceleration for deep learning tasks. 640 tensor cores accelerate tensor computation in deep. The software configuration of the experimental environment is as follows. The deep learning framework is PyTorch 1.1.0, and the versions of Python and Cuda are 3.7,10.0, respectively. some library files are as follows. numpy = ~1.19, scipy = ~1.0.1, future = ~0.16.0, matplotlib = ~2.2.2, pillow ≥ 6.2.0. opencv-python = ~4.6.0, scikit-image = ~0.14.0. Some of the deep learning parameters are as follows. learning Rate is 0.0001, Batch Size is 8. In summary, build appropriate hardware, operating systems, and other software platforms, providing a basic environment for realizing deep confrontation and backpropagation for image prediction of ancient ceramics.

Results

Pattern prediction results of missing area in ancient ceramic images based on deep adversarial

To show the prediction effect of the pattern prediction model for missing areas in ancient ceramic images. Comparative analysis is conducted between the iterative generation process of the missing area of the dragon-cloud pattern plate pattern, the missing pattern image, and the predicted pattern image. The model performance is analyzed in terms of accuracy, recall, and discriminator loss. Details are shown in Fig. 7.

Figure 7a shows the original image of the plate with auspicious dragon-cloud pattern. The image has not been mask-composited, and the missing portion of the pattern is shown as the color of the tattoo plate base blank itself, not covered by a white mask with a greyscale value of 255. Figure 7b shows a missing image of the dragon-cloud pattern plates. The original appearance of the dragon-cloud pattern inside the dish is maintained, and the circular outline of the outer rim of the dish is depicted by black lines. The missing portion is presented in a black grid and is located in the upper right corner of the disc. Figure 7c shows ground truth, created via Photoshop. Specifically, two layers are synthesized, the first layer is the original image and the second layer is the rotated image of the original image. The synthesis of the two layers fills the missing areas and the synthesized image is used as the ground truth. Figure 7d is the original image following the predictions. The artifact is extracted and the artifact of the missing part is predicted compared to the original image. Compared to ground truth, edge detection achieves the extraction of dragon-cloud disk artifact. In contrast to the pattern missing image, the pattern un-missing portion is aligned with the pattern missing image, and the pattern missing area is filled with predicted lines. Figure 7e shows the accuracy and recall curves of the pattern prediction model for the missing area in ancient ceramic images. At the beginning of the iteration (0–2000), the accuracy decreases rapidly and the recall increases rapidly, which are inversely related. Figure 7f shows the loss function curve of the pattern prediction model for the missing area in ancient ceramic images. That includes generator loss and discriminator loss. The loss of the discriminator to 0.689 starts with a fluctuation in the range of [0, 0.01], which gradually decreases with the number of iterations and finally converges to 0.5. The generator loss is a minimum loss of 0.56 over 40,000 iterations of steps. To sum up, the texture prediction model effectively reconstructs dragon-cloud patterns in ancient ceramics, demonstrating enhanced line clarity, complete structural continuity, and smooth curvature transitions. These restoration improvements directly translate to measurable performance gains, with heightened pattern recognition accuracy, improved recall rates, and reduced reconstruction loss compared to conventional restoration approaches.

Complementary prediction results of missing areas in ancient ceramic images based on reverse diffusion

To show the prediction effect of the complementary prediction model for missing areas in ancient ceramic images. Synthetic data, predict result and ground truth of both the blue-and-white mandarin duck lotus pond floral pattern plate and the blue-and-white lotus petal floral pattern plate are

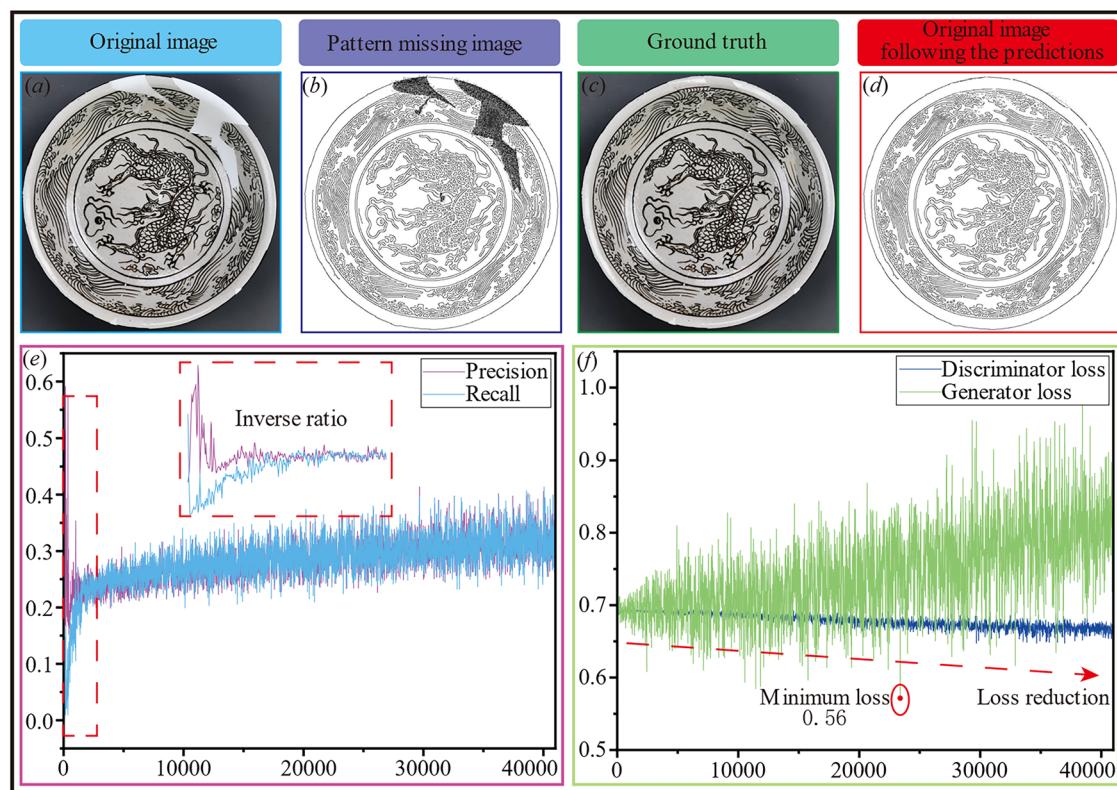


Fig. 7 | The pattern prediction result of missing area in ancient ceramic images based on deep adversarial. a Original image, **b** pattern missing image, **c** ground truth, **d** original image following the predictions, **e** the accuracy and recall curves of the pattern prediction model for the missing area in ancient ceramic images, **f** the loss function curve of the pattern prediction model for the missing area in ancient ceramic images.

compared and analyzed. Peak signal-to-noise ratio (PSNR), perceived image similarity (LPIPS), and mean square error (MSE) are analyzed. The details are shown in Fig. 8.

Figure 8a₁–a₃, respectively, show the synthetic data, predict result and ground truth of the blue-and-white mandarin duck lotus pond floral pattern plate. There are four missing areas of the stripe disc, with 3.02% of the area missing. Figure 8b₁–b₃ depict the synthetic data, predict result and ground truth of the blue lotus flower plate, respectively. There are 13 missing areas in the disc, among them, there are 11 missing areas of ornamentation, and there are 2 missing areas of the disc body. Figure 8b shows the PSNR radar plot. The PSNR increases from 7.91 to 7.97 over the course of 1000 to 10,000 iterations. Figure 8c shows the MSE and LPIPS metrics for this paper's algorithm (TM) (textual method) as well as the other methods. The smaller the value of MSE and LPIPS, the better the prediction. The MSEs for the NS method, the FMM, and the method of this paper (TM) are 0.3044, 0.2983, and 0.0676, respectively. The LPIPS are 0.2602, 0.2453, and 0.0699, respectively. The MSE as well as LPIPS of this paper's method is 0.2368, 0.1903 lower than the other two methods, respectively. Of these, MSE and LPIPS are explained in detail below. Mean square error (MSE) is a measure of the difference between the predicted value of the model and the actual observed value. A smaller MSE indicates a smaller difference between the predicted and actual values observed of the model, i.e., a better fit of the model. It is calculated as the square of the difference between the true value and the predicted value then summed and averaged. Learned perceptual image patch similarity (LPIPS) is a measure of image similarity that is based on high-level semantic features extracted by deep neural networks to compute the difference between two images. A lower value of LPIPS indicates that the two images are more similar and vice versa. Traditional methods rely on pixel-level differences for comparison and are unable to capture the characteristics of an image in the way the human visual system does. In contrast, LPIPS utilizes the responses of the layers within a pre-trained Convolutional Neural Network (CNN) as a feature representation

and constructs a perceptual loss function based on this. That is, LPIPS is more compatible with the human perceptual situation than traditional methods (e.g. L2/PSNR, SSIM, FSIM). Figure 8d shows the MSE and LPIPS between synthetic data, predict result and ground truth. Columns A and B represent the results for the blue and white mandarin duck and lotus pond floral dish and the blue and white lotus petal floral dish, respectively. In column A, the MSE and LPIPS generated from the synthetic data and the predicted results are 0.0303 and 0.0360, respectively. The MSE and LPIPS generated by ground truth and prediction results are 0.0137, 0.0166, respectively. The MSE and LPIPS generated by ground truth and prediction results are lower. Column B is the same. Lower MSE as well as LPIPS reflect biases associated with the use of synthetic data. The synthetic data contains missing areas, so the MSE and LPIPS are higher. To sum up, the complementary prediction model for missing areas in ancient ceramic images anticipate multi-scale deficiencies, including pattern and plate body loss, encompassing both outer rim and interior decorative details.

Virtual damage prediction results

To demonstrate the efficacy of the missing area prediction model for ancient ceramics, the actual damage caused by rotation and the virtual damage resulting from irregular figures are subjected to analysis. The details are shown in Fig. 9.

Figure 9a₁–e₁ shows the real damage image of rotation, and Fig. 9a₂–e₂ is the corresponding forecast result. The types of pattern plates include the dragon-cloud pattern plate, the blue-and-white mandarin duck lotus pond floral pattern plate, and the blue-and-white lotus petal floral pattern plate. The rotation angles and corresponding percentages of missing areas are 47.15° with 6.56%, 306.65° with 7.17%, 288.47° with 12.23%, 98.49° with 2.99%, and 72.75° with 11.97%, respectively. The missing area is represented as a white mask, and the prediction results fill the missing area with the corresponding pattern. Figure 9f₁–j₁ shows the virtual damage image of the irregular figure. Figure 9f₂–j₂ shows the corresponding prediction result.

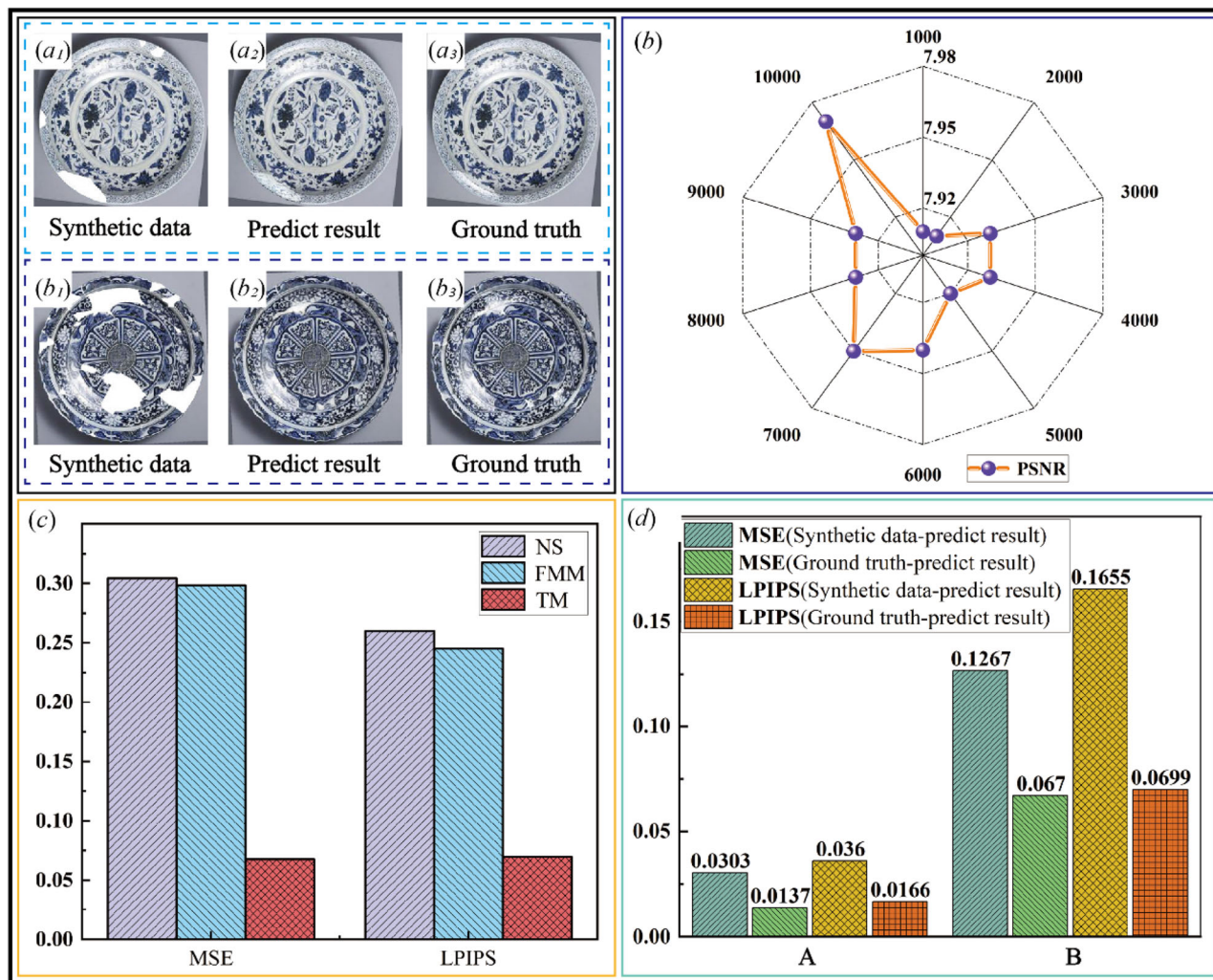


Fig. 8 | Complementary prediction results of the missing areas in ancient ceramic images. b The PSNR radar plot, **c** the MSE and LPIPS metrics for this paper's algorithm as well as the other methods, **d** the MSE and LPIPS between synthetic data, predict result and ground truth.

The proportions of rotation Angle and missing area were 276.06°, 5.36%, 239.76°, 13.66%, 121.58°, 2.78%, 157.62°, 1.14%, 277.05°, 25.35%, respectively. To sum up, the prediction model's robustness for ancient ceramic restoration is systematically validated through synthetic damage patterns encompassing diverse geometries, spatial configurations, and arbitrary rotational orientations. This confirms reliable prediction capabilities even when addressing compounded damage scenarios, including unavoidable secondary deterioration during conservation processes.

Discussion

To enable museums and restoration teams to apply the method of this paper to the restoration of different patterns, the implementation steps of the prediction method are established. Prediction consists of the following four main steps: image acquisition, dataset preparation, model training, and result test. The details are shown in Fig. 10.

Museums and restoration teams are able to use this digital prediction method in their process through the steps of image acquisition, dataset preparation, model training, and result testing. Images are captured by high-resolution cameras. The preparation of the dataset is achieved through Photoshop and MATLAB and mainly contains mask, synthesis, and ground truth images. In this paper, ground truth is achieved by "repetitive painting". That is, by copying the original image, creating a new layer, rotating the new layer, and finding the repetitive artifacts that correspond to the missing part. As shown above. In addition, museums and restoration teams can paint the

missing parts through "creative painting", making use of human initiative and the expertise of the restoration team. The digital prediction achieved by this method can provide a complete digital reconstruction of ancient ceramic patterns. Its value is not only reflected in the pattern of historical and cultural information preservation and secondary damage avoidance. It also extends to mural paintings, textiles, ancient architectural paintings, and other types of cultural heritage restoration scenarios. The method can reconstruct the topology of the surface ornamentation of cultural relics by 3D modeling technology, combined with the material database to match the historical process parameters. This can accurately compensate for physical damage (such as the symmetrical pattern reconstruction of Dunhuang wall paintings). It can also be used to simulate the aging characteristics of the material with the help of Generative Adversarial Networks (GAN) to complete the contactless virtual restoration. In addition, this method can preserve the historical traces in the restoration of stone cultural relics, and can also be based on reversible materials to achieve a full-fledged restoration. In the future, in the ancient book illustration restoration, lacquer decoration restoration, and other areas with greater potential for application, but need to build cultural relics category-specific digital restoration standards to avoid the risk of style inaccuracies.

(1) This paper proposes a digital prediction method for missing areas in ancient ceramic images based on deep adversarial and reverse diffusion. It contributes to the restoration of ancient ceramics with missing patterns due to natural disasters, time erosion, and other factors. The

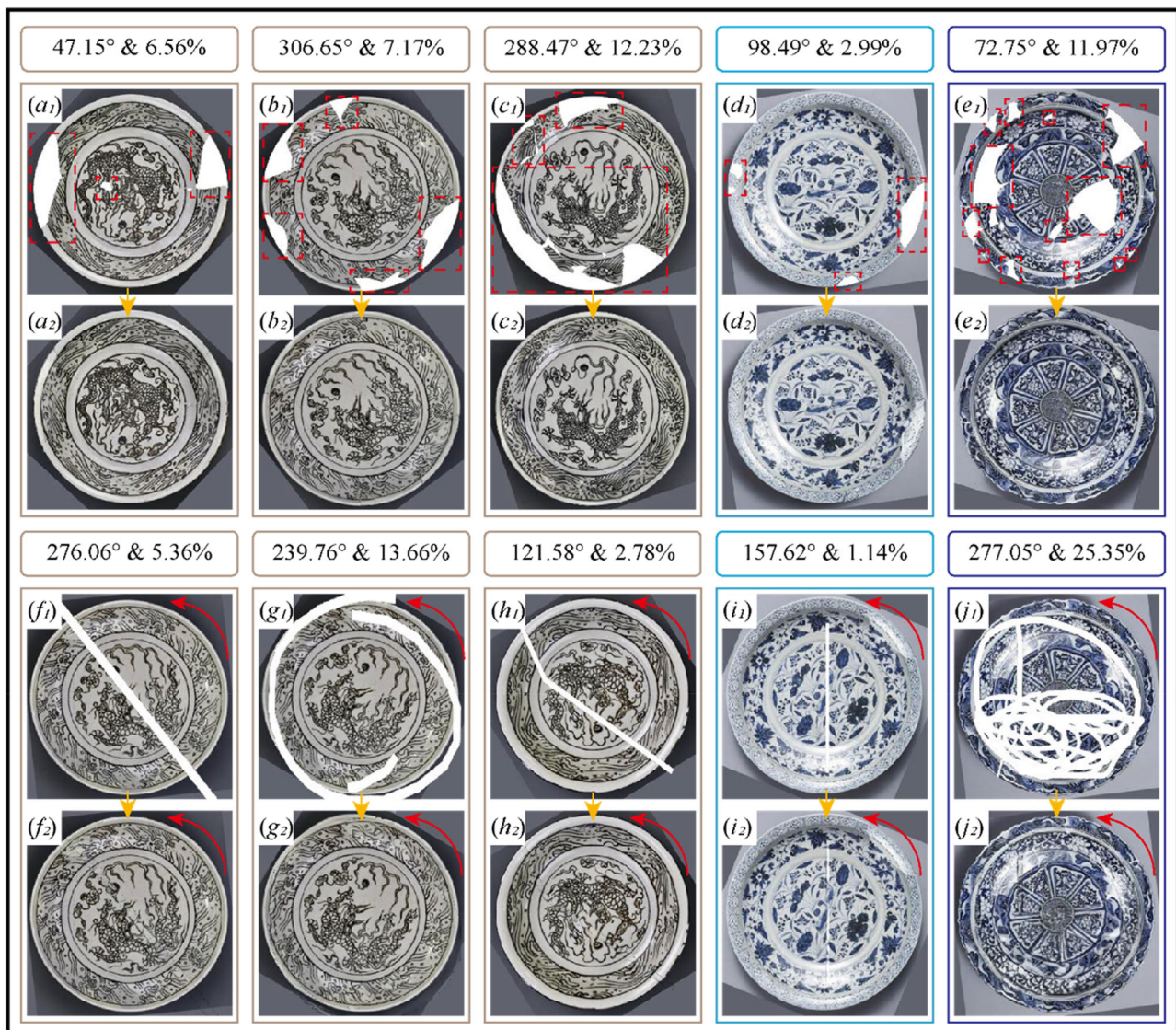


Fig. 9 | Prediction results of virtual damage in missing areas of ancient ceramic images.

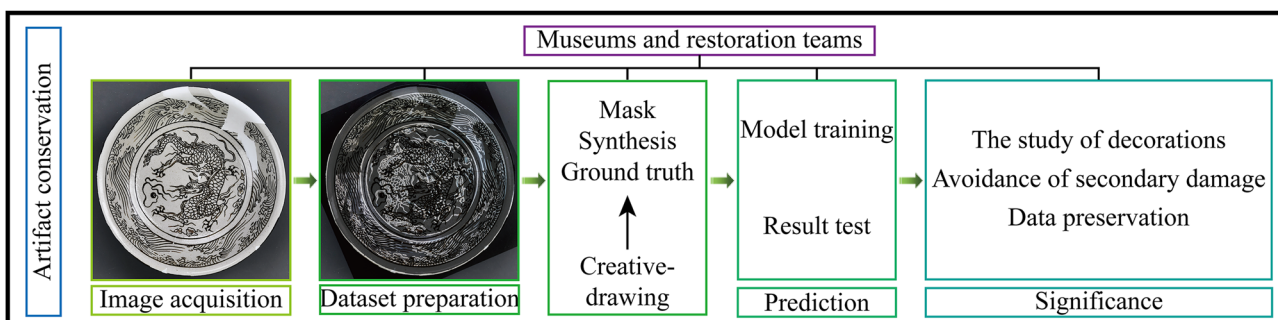


Fig. 10 | Flowchart for forecast realization.

experimental results show that the MSE and LPIPS of this paper's method are 0.0137 and 0.0166, respectively. In addition, the MSE and LPIPS of this paper's method are reduced by 0.2368 and 0.1903 compared to the FMM and NS methods, respectively. To sum up, the design of spectral normalization and local patch adversarial model is designed, in terms of technical practice. By constructing a hierarchical structure with a shift window mechanism, the pattern of missing areas in ancient ceramic images is predicted and complimented. In terms of

cultural significance, effective prediction for missing areas of ancient ceramic images is achieved. This contributes to the study of the historical background and cultural connotation of cultural artifacts, and the avoidance of secondary damage.

- (2) There are many types of ancient ceramics, such as bowls, jars, vases, stoves, pedestals, marigold holders, incense burners, high-footed cups, and other objects with complex shapes that are difficult to capture in a single image plane. The existing methods cannot achieve the prediction

of ancient ceramics with large curvature. However, 3D reconstruction techniques are important for working with complex shapes and designs. In future research, the combination of 3D reconstruction and digital prediction will be the main direction of attack. Extracting and converting the pattern of vases, jars, and other objects with complex shapes into 3D space. This breaks through the constraints of a single plane and enables prediction in 3D space.

Data availability

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation. Readers can obtain it by contacting the corresponding author's email.

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References

- Li, J. Semantic segmentation-based adaptive mining algorithm for ceramic cultural resource data. *Mob. Inf. Syst.* **2022**, 2815077 (2022).
- Criado-Boado, F. et al. Gazed pottery: an archaeometric-cognitive approach to material culture visuality. *J. Archaeol. Sci.* **154**, 105770 (2023).
- Heimann, R. B. On the nature of ceramics technology: from Empedocles to Dawkins. *Archaeometry* **67**, 55–71 (2025).
- VanValkenburgh, P., Kelloway, S. J., Privat, K. L., Sillar, B. & Quilter, J. Rethinking cultural hybridity and technology transfer: SEM microstructural analysis of lead glazed ceramics from early colonial Peru. *J. Archaeol. Sci.* **82**, 17–30 (2017).
- Zheng, B. et al. Image captioning for cultural artworks: a case study on ceramics. *Multimed. Syst.* **29**, 3223–3243 (2023).
- Limp, W. F. & Williamson, M. D. Strategies for international travel with “high-tech” archaeological field equipment. *Adv. Archaeol. Pract.* **5**, 382–387 (2017).
- Derech, N., Tal, A. & Shimshoni, I. Solving archaeological puzzles. *Pattern Recognit.* **119**, 108065 (2021).
- Guerneve, T., Subr, K. & Petillot, Y. Three-dimensional reconstruction of underwater objects using wide-aperture imaging SONAR. *J. Field Robot.* **35**, 890–905 (2018).
- Sha, G. et al. Mobile phone imaging point cloud modelling for 3D recording and measurement of small scenes of suspected fossil outcrops found in the field. *Geoheritage* **15**, 33 (2023).
- Lassandro, P., Fioriello, C. S., Lepore, M. & Zonno, M. Analysing, modelling and promoting tangible and intangible values of building heritage with historic flame lighting system. *J. Cult. Herit.* **47**, 166–179 (2021).
- Argyrou, A., Agapiou, A., Papakonstantinou, A. & Alexakis, D. D. Comparison of machine learning pixel-based classifiers for detecting archaeological ceramics. *Drones* **7**, 578 (2023).
- Fan, J. et al. Development, calibration, and image processing of underwater structured light vision system: a survey. *IEEE Trans. Instrum. Meas.* **72**, 1–18 (2023).
- Gao, Q., Long, T. & Zhou, Z. Mineral identification based on natural feature-oriented image processing and multi-label image classification. *Expert Syst. Appl.* **238**, 122111 (2024).
- Luo, L., Wang, X. & Guo, H. Transitioning from remote sensing archaeology to space archaeology: towards a paradigm shift. *Remote Sens. Environ.* **308**, 114200 (2024).
- Kokalj, Ž. et al. Machine learning-ready remote sensing data for Maya archaeology. *Sci. Data* **10**, 558 (2023).
- Küçükdemirci, M. & Sarris, A. GPR data processing and interpretation based on artificial intelligence approaches: Future perspectives for archaeological prospection. *Remote Sens.* **14**, 3377 (2022).
- Jin, X., Wang, X. & Xue, C. Nondestructive characterization and artificial intelligence recognition of acoustic identifiers of ancient ceramics. *Herit. Sci.* **11**, 144 (2023).
- Zhang, X., Zhai, D., Li, T., Zhou, Y. & Lin, Y. Image inpainting based on deep learning: a review. *Inf. Fusion* **90**, 74–94 (2023).
- Xiang, H. et al. Deep learning for image inpainting: a survey. *Pattern Recognit.* **134**, 109046 (2023).
- Wang, Q., Wang, Z., Zhang, X. & Feng, G. Art image inpainting with style-guided dual-branch inpainting network. *IEEE Trans. Multimed.* **26**, 8026–8037 (2024).
- Liao, J., Zhang, J., Ye, L. & Wei, X. Cross-attention and cycle-consistency-based haptic to image inpainting. *IEEE Signal Process. Lett.* **31**, 1650–1654 (2024).
- Ma, X. et al. Uncertainty-aware image inpainting with adaptive feedback network. *Expert Syst. Appl.* **235**, 121148 (2024).
- Wan, W. & Liu, J. Nonlocal patches based Gaussian mixture model for image inpainting. *Appl. Math. Model.* **87**, 317–331 (2020).

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Author contributions

Dahai Liao: Conceptualization, methodology, writing—original draft; Tian Zeng: Data curation, writing—original draft, visualization; Jianfei Yang: Software, writing—original draft; Guoxing Tang: Supervision and investigation; Xiaoling Zhou: Writing—review and editing validation. Zhijuan Deng: Conceptualization, resources, and supervision.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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